The mass of the electron as an electromagnetic effect

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Take a look at the equation for the magnetic flux around a moving electron in the nonrelativistic case, the simple Biot-Savarts law

$$\boxed{\bar{B} = \frac{\mu_0 e}{4\pi} \frac{\bar{v} \times \bar{r}}{r^3} \qquad (1)}$$

The energy-density of this field is according to eq. (2)

$$\boxed{u = \frac{\bar{B}^2}{2\mu_0} \qquad (\bar{B}^2 = \bar{B} \cdot \bar{B}) \quad (2)}$$

In order to calculate the magnetic energy of the moving electron we insert (1) into (2) to get (3).

$$\boxed{u = \frac{\mu_0 e^2}{32\pi^2} \frac{v^2 \sin^2\theta}{r^4} \qquad (3)}$$

Lets integrate eq. (3) outside the electron to find out the entire energy of its magnetic field.

$$\boxed{U = \frac{\mu_0 e^2}{12\pi} \frac{v^2}{r_e} \qquad (4)}$$

Lets install the classical electron radius r_e,

$$\boxed{r_e = \frac{e^2}{4\pi\,\varepsilon_0 c^2 m_e} = \frac{\mu_0 e^2}{4\pi\, m_e} \qquad (5)}$$

in (4) to get

$$\boxed{U = \frac{m_e v^2}{3} \qquad (6)}$$

This reminds very much of the equation of the kinetic energy of the electron. Only 1/6 of the kinetic energy is missing but remember that I haven't included the field inside the electron in the calculus. Can it be so simple that the inertia of the electron is due to Lenz' law?